

Obstacle Avoidance Guidance for Mars Powered Descent Using Convex Optimization and Elevation Angle

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인하대학교 항공우주제어시스템연구실

학부연구생 변 정 우



인하대학교
INHA UNIVERSITY



항공우주제어시스템연구실
Aerospace Control & Systems Laboratory

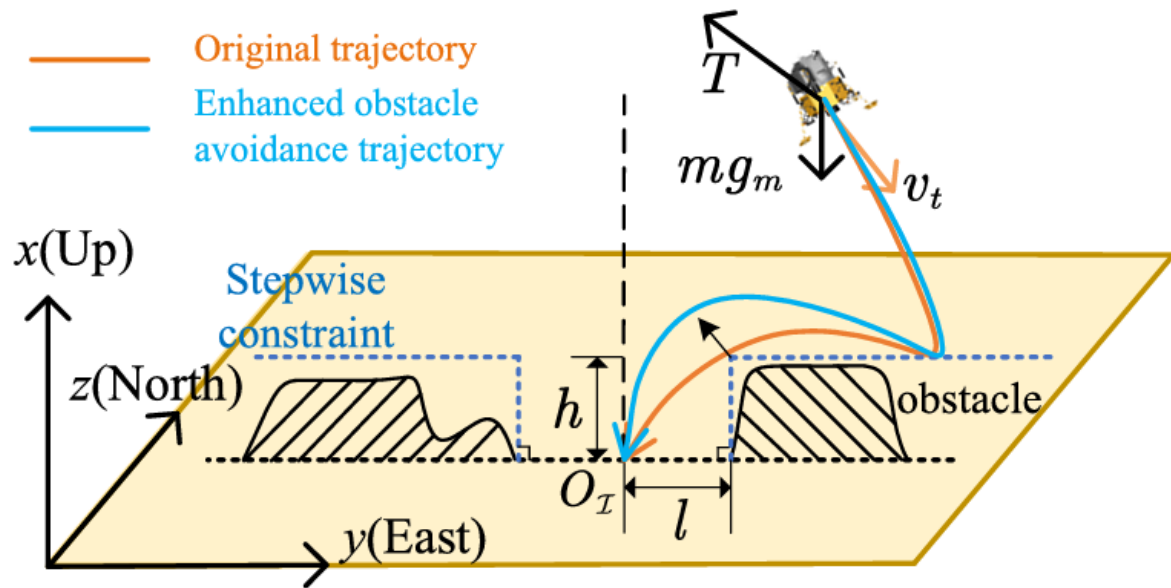
- **Obstacle Avoidance Guidance for Mars Powered Descent Using Convex Optimization and Elevation Angle**
 - Acta Astronautica 248 (2026) 296 – 313
 - Authors: Duozhi Gao, Youmin Gong, Yanning Guo, Yizheng Xiao, Edoardo Fadda, Paolo Brandimarte
- **Methods**
 - Algorithm 1: improved convexification + fuel-optimal obstacle avoidance
 - Algorithm 2: 선택한 fuel, time에서 vertical-biased trajectory 생성
- **Applications**
 - Mars powered descent, obstacle avoidance, terminal camera field-of-view improvement

1. Introduction
2. Problem formulation
3. Improved lossless convexification and obstacle avoidance
4. Elevation-angle-based powered descent guidance
5. Numerical experiments
6. Conclusion

■ 최근 Mars powered descent 동향

- 과거 landing 미션은 평탄하고 장애물이 적은 landing site를 선호
- powered descent: 제한된 thrust와 fuel 내에서 감속
- 최근 관심사는 crater, canyon, volcanic region 등 복잡한 지형을 요구
- 단순 obstacle avoidance를 넘어 terminal camera FoV와 vertical landing 특성까지 고려할 필요

→ 목표: convex optimization의 계산 효율 유지 + complex obstacle constraints 처리+ FoV 향상



■ 기존 guidance 방법의 한계

- Analytical guidance: thrust constraint, parameter sensitivity, fuel optimality에 한계
- Learning-based guidance: 계산은 빠르지만 stability&reliability 문제가 남음
- Optimization-based guidance: constraint handling과 fuel minimization에 강하지만, non-convex 계산 부담
- 기존 convex 접근에서는 장애물 제약 처리와, 연료 최적화 이후의 접근 형상 개선을 별도로 다룰 필요가 있음

■ Contributions

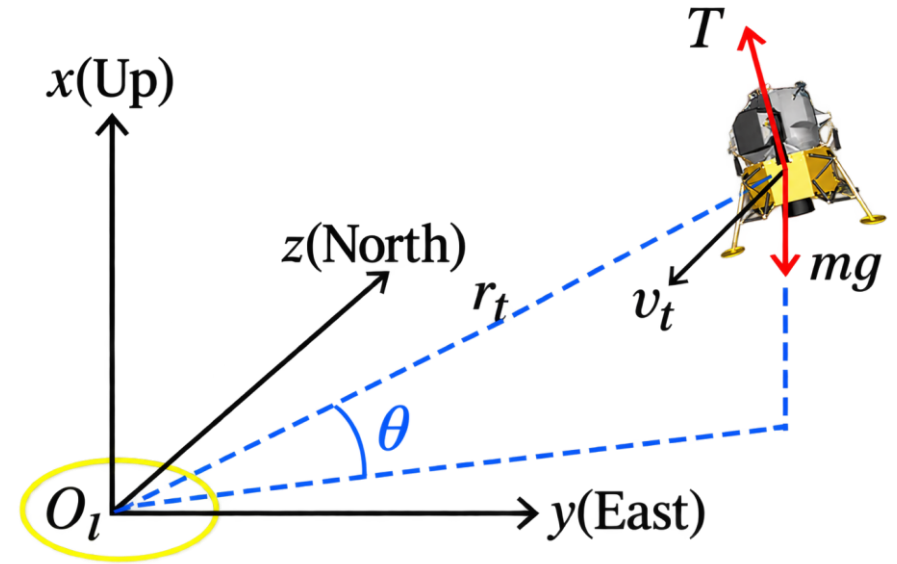
- Iterative Taylor update + Homotopy based sequential convex framework:
thrust 근사 개선 및 relaxed glide slope / stepwise obstacle constraints 처리
- Axis weighted position integral을 objective로 도입 → trajectory를 vertical direction으로 유도
- Elevation angle integral을 기준으로 optimal terminal time과 obstacle avoidance trajectory 결정

▪ Powered descent 3-DoF mathematical modeling

- State: position $\mathbf{r}$, velocity $\mathbf{v}$, mass m
- control: thrust vector $\mathbf{T}$
- Axes: x upward (수직 방향), y east, z north
- Constant gravity / atmosphere와 planetary rotation은 제외

▪ Elevation angle θ

- landing site 원점 기준으로 lander 위치벡터가 수평면과 이루는 각 θ
- 고도 r_x 가 커지거나 수평거리 $\sqrt{r_y^2 + r_z^2}$ 가 작아질수록 θ 가 증가
- 큰 elevation angle은 terminal camera FoV 확보와 vertical approach에 유리



$$\dot{\mathbf{r}}(t) = \mathbf{v}(t)$$

$$\dot{\mathbf{v}}(t) = \mathbf{g}_m + \frac{\mathbf{T}(t)}{m(t)}$$

$$\dot{m}(t) = -\lambda \|\mathbf{T}(t)\|_2$$

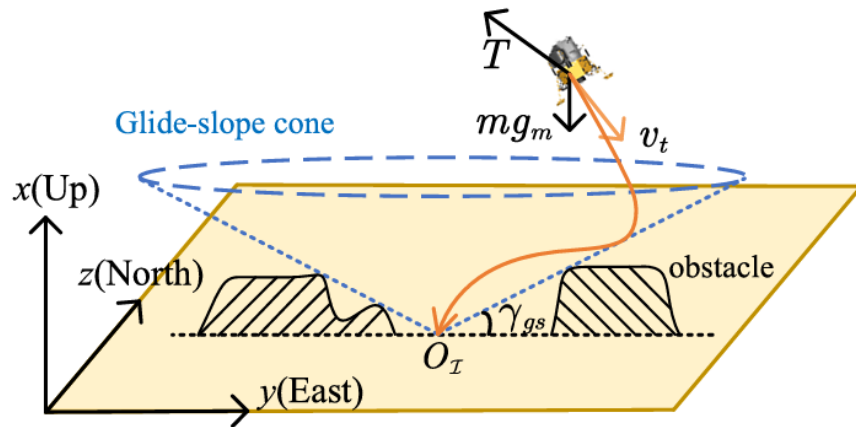
$$\theta = \arcsin\left(\frac{r_x}{\sqrt{r_x^2 + r_y^2 + r_z^2}}\right) = \arctan\left(\frac{r_x}{\sqrt{r_y^2 + r_z^2}}\right)$$

■ Obstacle constraint formulation

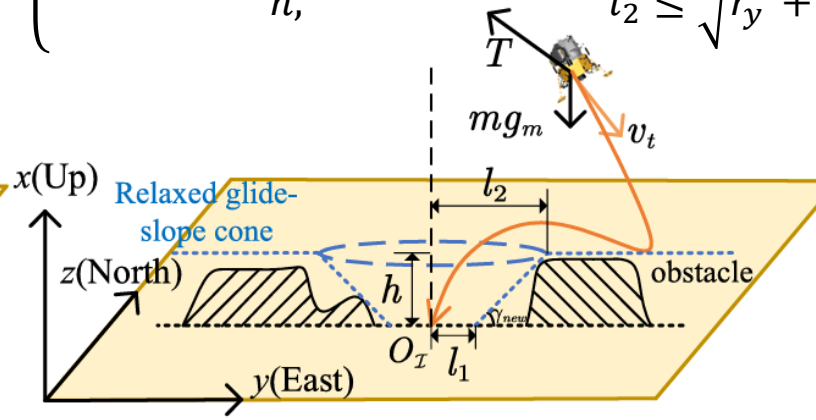
- conventional glide-slope: landing site 주변의 conical safe region을 보존
- relaxed glide-slope: 충분한 altitude에서 constraint를 완화
- stepwise constraint: safe radius 밖에서 safe altitude를 요구

$$\tan \gamma_{gs} \leq \frac{r_x}{\sqrt{r_y^2 + r_z^2}}$$

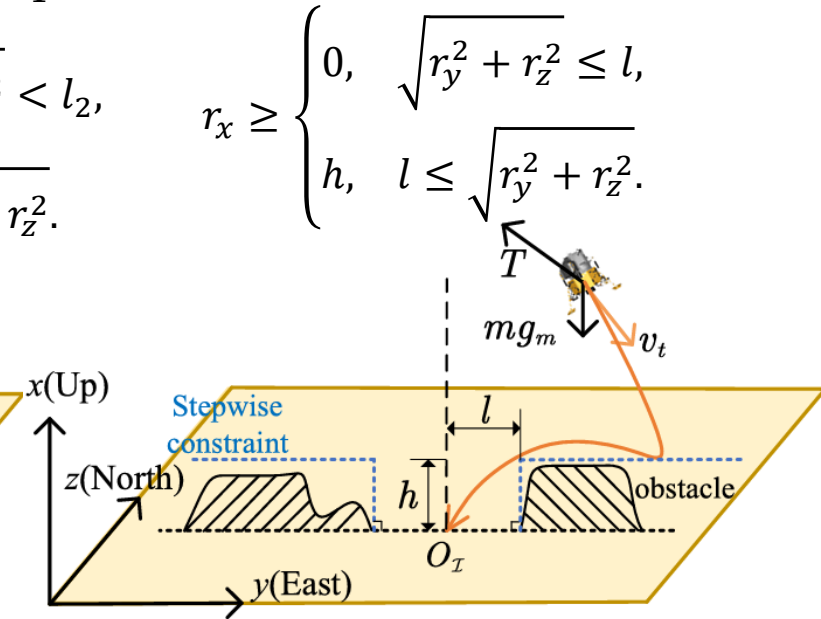
$$r_x \geq \begin{cases} 0, & \sqrt{r_y^2 + r_z^2} \leq l_1, \\ \left(\sqrt{r_y^2 + r_z^2} - l_1 \right) \tan \gamma_{new}, & l_1 < \sqrt{r_y^2 + r_z^2} < l_2, \\ h, & l_2 \leq \sqrt{r_y^2 + r_z^2}. \end{cases}$$



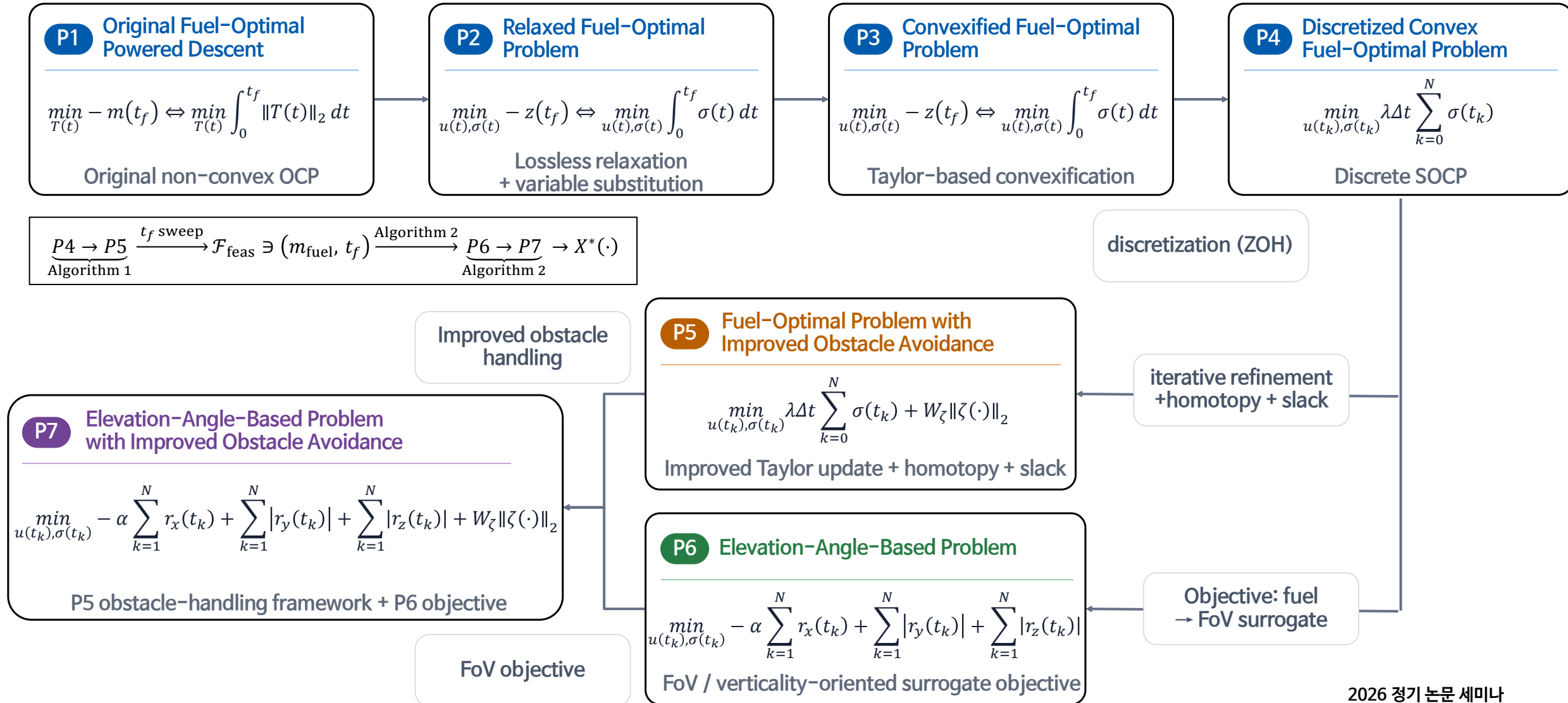
(a) Illustration of conventional glide-slope constraint.



(b) Illustration of relaxed glide-slope constraint.



(c) Illustration of stepwise constraint.



■ Problem 1: Fuel-optimal powered descent

- Final mass maximization과 thrust magnitude integral minimization을 동등한 fuel-optimal objective로 정의
- Dynamics, boundary conditions, thrust magnitude + obstacle avoidance constraints를 동시에 고려
- Minimum thrust와 mass-dependent dynamics로 인해 original problem은 non-convex OCP

$$\min_{T(t)} -m(t_f) \Leftrightarrow \min_{T(t)} \int_0^{t_f} \|T(t)\|_2 dt$$

[Objective Function]

$$\begin{aligned} \dot{r} &= v \\ \dot{v} &= g_m + \frac{T(t)}{m(t)} \\ \dot{m}(t) &= -\lambda \|T(t)\|_2 \end{aligned}$$

$$\begin{aligned} r(t_0) &= r_0, v(t_0) = v_0, m(t_0) = m_{wet} \\ r(t_f) &= r_f, v(t_f) = v_f, m(t_f) \geq m_{dry} \end{aligned}$$

[Dynamics & Boundary Condition]

$$\begin{aligned} 0 < T_{min} \leq \|T(t)\|_2 \leq T_{max} \\ \tan \gamma_{gs} &\leq \frac{r_x}{\sqrt{r_y^2 + r_z^2}} \end{aligned}$$

[Thrust & Obstacle Constraints]

■ Problem 2: Relaxed fuel-optimal powered descent

- 추력 크기의 스칼라 보조변수 Γ 를 도입하여 $\|T(t)\|_2 \leq \Gamma(t)$ 로 완화하고, $\dot{m} = -\lambda\Gamma(t)$ 로 표현
(Slack variable과 $z = \ln m$, $u = \frac{T}{m}$, $\sigma = \frac{\Gamma}{m}$ 변수 도입)
- Mass division을 제거하여 dynamics와 fuel objective를 convex formulation에 적합한 형태로 변환
- 변환 이후에도 exponential thrust bounds의 non-convex가 남아 있음

$$\min_{u(t), \sigma(t)} -z(t_f) \Leftrightarrow \min_{u(t), \sigma(t)} \int_0^{t_f} \sigma(t) dt$$

[Objective Function]

$$\begin{aligned} \dot{r} &= v, & \dot{v} &= g_m + u \\ \dot{z} &= -\lambda\sigma, & \|u\|_2 &\leq \sigma \\ r(t_0) &= r_0, & v(t_0) &= v_0 \\ z(t_0) &= \ln m_{wet} \\ r(t_f) &= r_f, & v(t_f) &= v_f \\ z(t_f) &\geq \ln m_{dry} \end{aligned}$$

[Relaxed Dynamics & Boundary Conditions]

$$\begin{aligned} 0 &< \frac{T_{min}}{m(t)} \leq \sigma(t) \leq \frac{T_{max}}{m(t)} \\ T_{min}e^{-z(t)} &\leq \sigma(t) \leq T_{max}e^{-z(t)} \\ \tan\gamma_{gs} &\leq \frac{r_x}{\sqrt{r_y^2 + r_z^2}} \end{aligned}$$

[Thrust & Glide-Slope Constraints]

■ Problem 3: Convexified fuel-optimal powered descent

- Lower thrust bound: $T_{\min} e^{-z} \leq \sigma$ 를 convex quadratic 형태로 근사하여 SOCP에 맞춤
- Upper thrust bound: $\sigma \leq T_{\max} e^{-z}$ 를 affine 형태로 근사하여 non-convexity를 처리
- Dynamics와 boundary conditions는 유지하고, Taylor-approximated continuous convex problem을 구성

$$\min_{u(t), \sigma(t)} -z(t_f) \Leftrightarrow \min_{u(t), \sigma(t)} \int_0^{t_f} \sigma(t) dt$$

$$z_l(t) = \ln(m_{\text{wet}} - \lambda T_{\min} t)$$

$$z_u(t) = \ln(m_{\text{wet}} - \lambda T_{\max} t)$$

[Objective Function & Mass References]

$$\dot{r} = v, \quad \dot{v} = g_m + u$$

$$\dot{z} = -\lambda \sigma, \quad \|u\|_2 \leq \sigma$$

$$r(t_0) = r_0, \quad v(t_0) = v_0$$

$$z(t_0) = \ln m_{\text{wet}}$$

$$r(t_f) = r_f, \quad v(t_f) = v_f$$

$$z(t_f) \geq \ln m_{\text{dry}}$$

[Relaxed Dynamics & Boundary Conditions]

$$d_l = z - z_l, \quad d_u = z - z_u$$

$$T_{\min} e^{-z_l} \left[1 - d_l + \frac{1}{2} d_l^2 \right] \leq \sigma$$

$$\sigma \leq T_{\max} e^{-z_u} [1 - d_u]$$

$$\sqrt{r_y^2 + r_z^2} \tan \gamma_{gs} \leq r_x$$

[Convexified Thrust
& Glide-Slope Constraints]

■ Problem 4: Discretized and convexified fuel-optimal powered descent

- Continuous convex formulation을 zero-order hold 기반 discrete-time system으로 변환
- Dynamics, boundary, thrust 및 glide-slope constraints를 각 discretization node에서 적용
- CVX: 모델링 + MOSEK: convex solver로 직접 해결 가능한 discrete SOCP

$$\min_{u_k, \sigma_k} -z_{N+1} \Leftrightarrow \min_{u_k, \sigma_k} \lambda \Delta t \sum_{k=0}^N \sigma_k$$

$$t_k = k\Delta t, \quad \Delta t = \frac{t_f - t_0}{N+1}$$

$$z_{l,k} = \ln(m_{wet} - \lambda T_{min} t_k)$$

$$z_{u,k} = \ln(m_{wet} - \lambda T_{max} t_k)$$

$$q_k = q(t_k)$$

[Objective Function & Discrete Definition]

$$X_{k+1} = A_d X_k + B_d [u_k^T \sigma_k]^T + B_d G_m$$

$$X_k = [r_k^T \ v_k^T \ z_k]^T$$

$$X_0 = [r_0^T \ v_0^T \ \ln m_{wet}]^T$$

$$r_{N+1} = r_f, \quad v_{N+1} = v_f$$

$$z_{N+1} \geq \ln m_{dry}, \quad \|u_k\|_2 \leq \sigma_k$$

[Discrete Dynamics & Boundary Conditions]

$$T_{min} e^{-z_{l,k}} \left[1 - d_{l,k} + \frac{1}{2} d_{l,k}^2 \right] \leq \sigma_k$$

$$\sigma_k \leq T_{max} e^{-z_{u,k}} [1 - d_{u,k}]$$

$$d_{l,k} = z_k - z_{l,k}, \quad d_{u,k} = z_k - z_{u,k}$$

$$\sqrt{r_{y,k}^2 + r_{z,k}^2} \tan \gamma_{gs} \leq r_{x,k}$$

[Convexified Thrust
& Glide-Slope Constraints]

■ Problem 5: Discrete fuel-optimal powered descent with obstacle avoidance

- Previous solution의 log-mass trajectory를 새로운 Taylor expansion point로 반복 갱신
- Slack variable로 intermediate constraint violation을 허용하고, penalty로 slack 사용을 억제
- Homotopy parameter를 증가시키며 relaxed glide-slope / stepwise obstacle constraint를 점진적으로 적용

$$\min_{u_k, \sigma_k} \left[\lambda \Delta t \sum_{k=0}^N \sigma_k + W_{\zeta} \|\zeta(\cdot)\|_2 \right]$$

$$d_k = z_k - \tilde{z}_k, \quad \tilde{z}_k = \tilde{z}(t_k)$$

[Objective Function & Slack Penalty]

$$X_{k+1} = A_d X_k + B_d [u_k^T \sigma_k]^T + B_d G_m$$

$$X_k = [r_k^T v_k^T z_k]^T$$

$$X_0 = [r_0^T v_0^T \ln m_{wet}]^T$$

$$r_{N+1} = r_f, \quad v_{N+1} = v_f$$

$$z_{N+1} \geq \ln m_{dry}, \quad \|u_k\|_2 \leq \sigma_k$$

[Discrete Dynamics & Boundary Conditions]

$$T_{min} e^{-\tilde{z}_k} \left[1 - d_k + \frac{1}{2} d_k^2 \right] \leq \sigma_k$$

$$\sigma_k \leq T_{max} e^{-\tilde{z}_k} (1 - d_k)$$

$$r_{x,k} \geq F_k + \zeta_k, \quad 0 \leq \delta \leq 1$$

$$F_k = F(X_k, \delta)$$

$$F_k = \begin{cases} 0, & \rho_k \leq l_1 \\ \delta(\rho_k - l_1) \tan \gamma_{new}, & l_1 < \rho_k < l_2 \\ \delta h, & l_2 \leq \rho_k \end{cases}$$

$$\text{or } F_k = \begin{cases} 0, & \rho_k \leq l \\ \delta h, & l \leq \rho_k \end{cases}$$

$$\rho_k = \sqrt{r_{y,k}^2 + r_{z,k}^2}, \quad \tan \gamma_{new} = \frac{h}{l_2 - l_1}$$

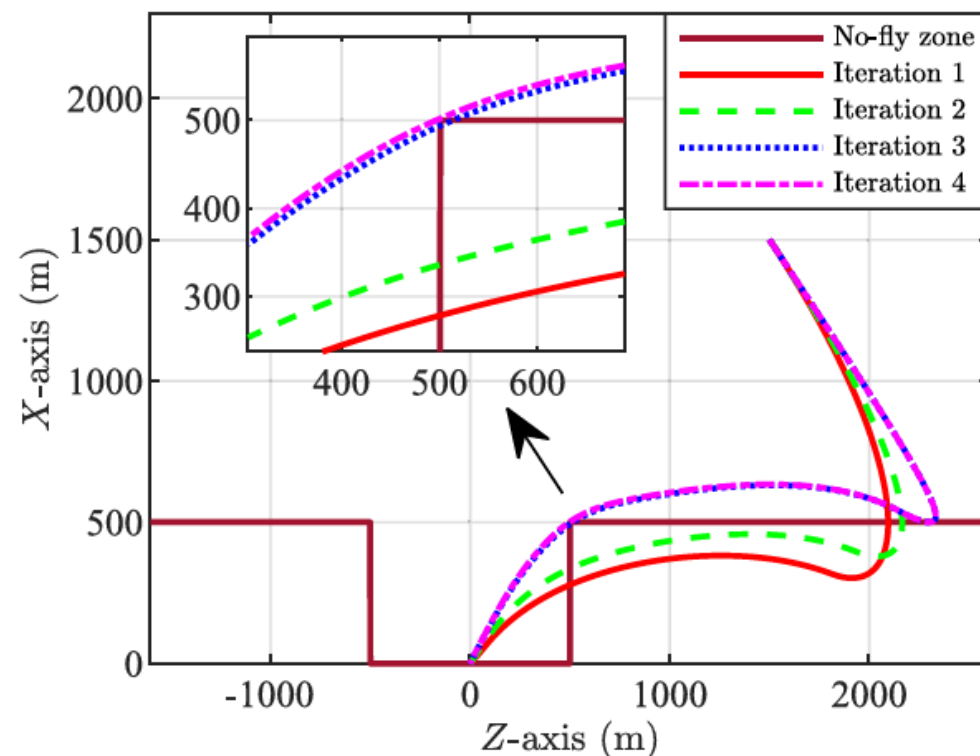
[Thrust & Homotopy Constraints]

Homotopy iteration

- Relaxed / stepwise obstacle constraints는 nonconvex 분할 구조이므로 직접적인 SOCP 적용이 어려움
- Reference 기반 장애물 제약을 한 번에 적용하면 thrust 제한과 함께 만족시키기 어려울 수 있음
- 단순한 초기 문제에서 시작하여 $\delta: 0 \rightarrow 1$ 로 원래 장애물 제약을 점진적으로 적용
- Slack으로 중간 제약 위반을 허용하면서 궤적을 갱신하고, 최종적으로 원래 constraints의 만족 여부를 확인

$$r_x(t_k) \geq F(\mathbf{X}(t_k), \delta) + \zeta(t_k), \quad \delta = \frac{\min(j, H)}{H}$$

$$\min \lambda \Delta t \sum_{k=0}^N \sigma(t_k) + W_\zeta \|\zeta(\cdot)\|_2$$



Algorithm 1 Improved lossless convexification algorithm with obstacle avoidance for fuel optimal powered descent.

```
1: Define tolerance  $\varepsilon_\zeta > 0$ . Define homotopy iteration count  $H \in \mathbb{Z}^+$ .
2: Define iteration count  $j = 0$ . Define Endflag=false.
3: Initial trajectory: solve Problem 4 with  $\gamma_{gs} = 0$  to obtain  $\tilde{u}(\cdot), \tilde{\sigma}(\cdot), \tilde{X}(\cdot)$ .
4: while Endflag=false do
5:    $j = j + 1$ .
6:   Assign  $\delta = \min(j, H) / H$ .
7:   Calculate  $F(\tilde{X}(t_k), \delta)$  according to Eq. (32) or (33).
8:   Assign  $z_l(t_k) = \tilde{z}(t_k), z_u(t_k) = \tilde{z}(t_k)$  according to Eq. (30).
9:   Solve Problem 5. Denote the solution as  $\hat{u}(\cdot), \hat{\sigma}(\cdot), \hat{X}(\cdot)$ .
10:  Compute the norm of the slack variable  $\|\zeta(\cdot)\|_2$ .
11:  Check whether the obtained solution satisfies all constraints.
12:  if All constraints satisfied and  $\|\zeta(\cdot)\|_2 \leq \varepsilon_\zeta$  and  $j \geq (H + 1)$  then
13:    Endflag=true.
14:  else
15:    Set the solution  $\hat{u}(\cdot), \hat{\sigma}(\cdot), \hat{X}(\cdot)$  as the initial trajectory. Assign
     $\tilde{u}(\cdot) = \hat{u}(\cdot), \tilde{\sigma}(\cdot) = \hat{\sigma}(\cdot), \tilde{X}(\cdot) = \hat{X}(\cdot)$ .
16:  Endflag=false.
17: end if
18: end while
19: Assign  $u^*(\cdot) = \hat{u}(\cdot), \sigma^*(\cdot) = \hat{\sigma}(\cdot), X^*(\cdot) = \hat{X}(\cdot)$ .
20: return  $u^*(\cdot), \sigma^*(\cdot), X^*(\cdot)$  as the fuel optimal solution with obstacle avoidance.
```

0) Problem 1~3

1) 초기 설정 및 초기 궤적 생성
: Problem4

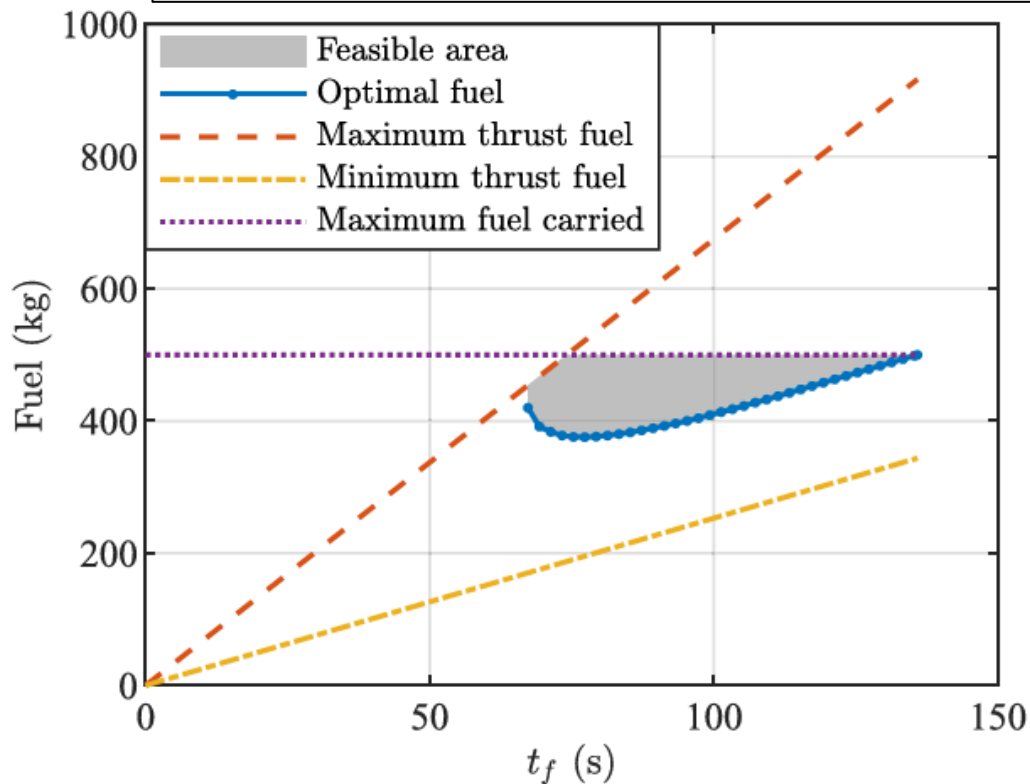
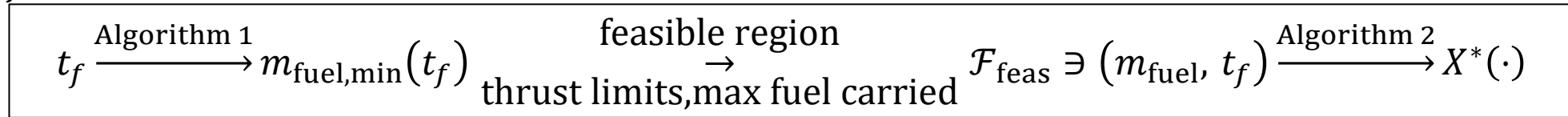
2) Taylor / Homotopy update → SOCP 해결
: Problem 5

3) 수렴 판정 / 미수렴 시 갱신
: 모든 constraints, $\|\zeta(\cdot)\|_2 \leq \varepsilon_\zeta, j \geq H + 1$ 을 만족하면 종료
미충족 시 현재 해를 reference로 갱신하여 반복

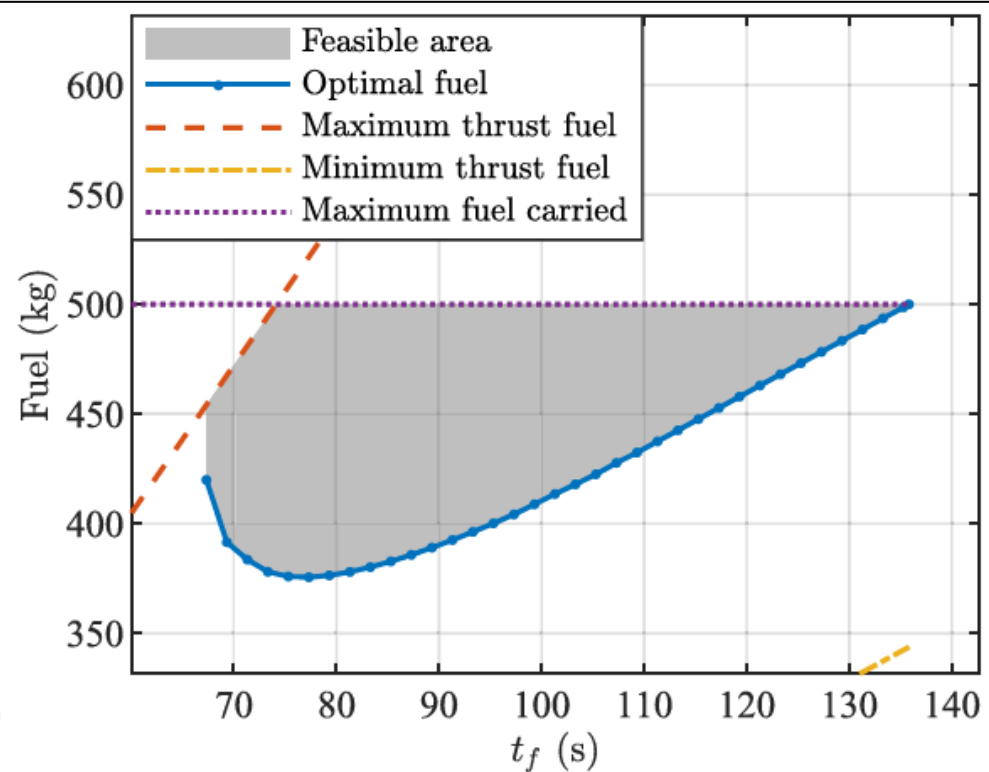
4) 최종 fuel-optimal 해 반환

Feasible fuel-time area과 초기 설정

- Algorithm 1로 feasible fuel - time region을 구성하고, 내부의 (m_{fuel}, t_f) 를 Algorithm 2의 입력으로 선택 (전체 t_f 중 하나 선택 / 필요한 최소 연료량 계산 → 이 pair들의 나열: Optimal fuel 점선)



(a) Global diagram of feasible area.



(b) Localized enlargement of feasible region.

■ Problem 6: Elevation-angle-based descent with basic obstacle avoidance

- 주어진 fuel budget과 terminal time에서, 연료 최소화 대신 **axis-weighted position objective**를 최소화
- Altitude**를 증가시키고 **horizontal distance**를 감소시켜 elevation angle과 terminal FoV를 간접적으로 개선
- Conventional glide-slope를 사용하는 basic obstacle-avoidance formulation으로 구성

$$\min_{u_k, \sigma_k} \sum_{k=1}^N (-\alpha r_{x,k} + |r_{y,k}| + |r_{z,k}|)$$

$$m_{dry} = m_{wet} - m_{fuel}$$

$$t_k = k\Delta t, \quad \Delta t = \frac{t_f - t_0}{N+1}$$

$$z_{l,k} = \ln(m_{wet} - \lambda T_{min} t_k)$$

$$z_{u,k} = \ln(m_{wet} - \lambda T_{max} t_k)$$

$$q_k = q(t_k)$$

[Axis-Weighted Objective & Fuel Budget]

$$X_{k+1} = A_d X_k + B_d [u_k^T \sigma_k]^T + B_d G_m$$

$$X_k = [r_k^T v_k^T z_k]^T$$

$$X_0 = [r_0^T v_0^T \ln m_{wet}]^T$$

$$r_{N+1} = r_f, \quad v_{N+1} = v_f$$

$$z_{N+1} \geq \ln m_{dry}, \quad \|u_k\|_2 \leq \sigma_k$$

[Discrete Dynamics & Boundary Conditions]

$$T_{min} e^{-z_{l,k}} \left[1 - d_{l,k} + \frac{1}{2} d_{l,k}^2 \right] \leq \sigma_k$$

$$\sigma_k \leq T_{max} e^{-z_{u,k}} [1 - d_{u,k}]$$

$$d_{l,k} = z_k - z_{l,k}, \quad d_{u,k} = z_k - z_{u,k}$$

$$\sqrt{r_{y,k}^2 + r_{z,k}^2} \tan \gamma_{gs} \leq r_{x,k}$$

[Convexified Thrust
& Glide-Slope Constraints]

■ Effect of Vertical-Bias Coefficient α

- α 는 고도 증가를 보상하는 가중치로, trajectory의 vertical bias를 조절
- 작은 α : 회피 구간의 고도 확보가 제한되어 장애물 경계에 가까운 궤적이 생성
- 큰 α : 초기 고도 확보에 치우쳐 terminal phase의 궤적 조정이 제한
- * 해당 논문은 $\alpha \in [0.5, 2]$ 를 적절한 범위로 제안하고, 기본값으로 $\alpha = 1$ 을 사용

$$\min \int_{t_0}^{t_f} (-\alpha r_x(t) + |r_y(t)| + |r_z(t)|) dt$$

Stepwise constraint

$$t_f = 80 \text{ s}$$

$$\text{fuel} = 400 \text{ kg}$$

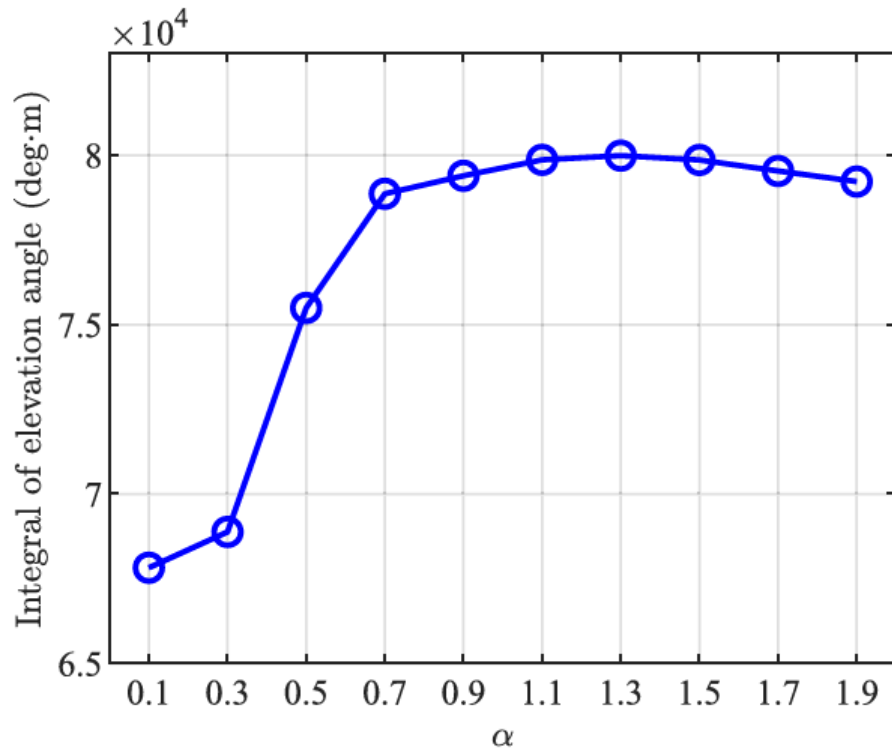


Fig. 8

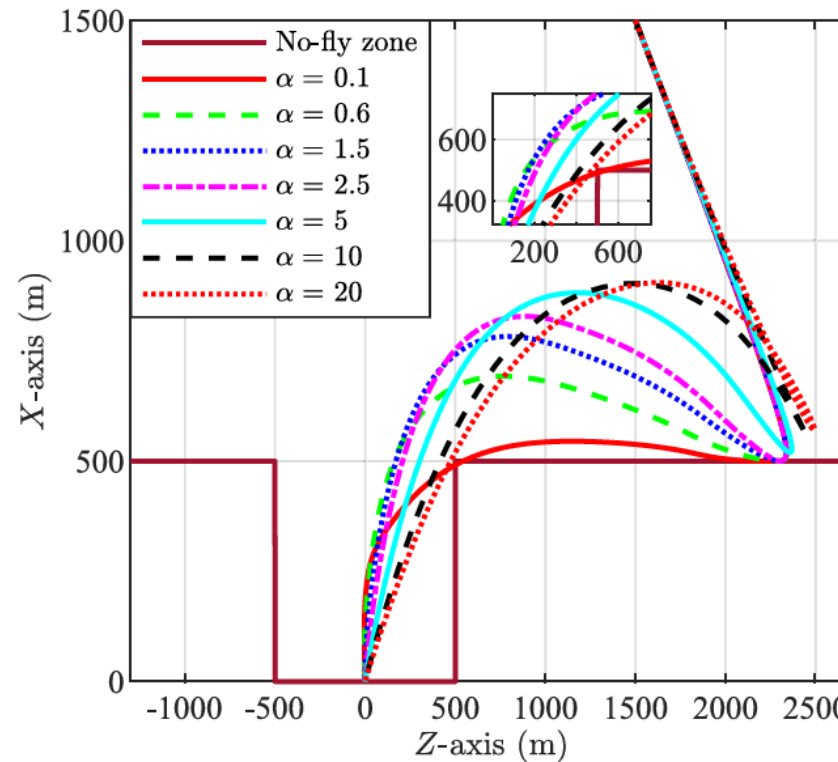


Fig. 9

- **Problem 7: Elevation-angle-based descent with improved obstacle avoidance**
 - Problem 6의 **FoV-oriented objective**와 Problem 5의 **improved obstacle-handling framework**를 결합
 - Homotopy, iterative Taylor update, slack relaxation을 이용해 relaxed / stepwise constraints를 처리
 - 주어진 fuel budget과 terminal time에서, P5의 반복 구조로 회피 제약을 처리하며 vertical-biased trajectory를 계산

$$\min_{u_k, \sigma_k} \sum_{k=1}^N \left(-\alpha r_{x,k} + |r_{y,k}| + |r_{z,k}| \right) + W_\zeta \| \zeta(\cdot) \|_2$$

$$T_{min} e^{-\tilde{z}_k} \left[1 - d_k + \frac{1}{2} d_k^2 \right] \leq \sigma_k$$

$$\sigma_k \leq T_{max} e^{-\tilde{z}_k} (1 - d_k)$$

$$d_k = z_k - \tilde{z}_k, \quad \tilde{z}_k = \tilde{z}(t_k)$$

$$m_{dry} = m_{wet} - m_{fuel}$$

[Objective & Fuel Budget]

$$X_{k+1} = A_d X_k + B_d [u_k^T \sigma_k]^T + B_d G_m$$

$$X_k = [r_k^T v_k^T z_k]^T$$

$$X_0 = [r_0^T v_0^T \ln m_{wet}]^T$$

$$r_{N+1} = r_f, \quad v_{N+1} = v_f$$

$$z_{N+1} \geq \ln m_{dry}, \quad \| u_k \|_2 \leq \sigma_k$$

[Discrete Dynamics & Boundary Conditions]

$$r_{x,k} \geq F_k + \zeta_k, \quad 0 \leq \delta \leq 1$$

$$F_k = F(\tilde{X}_k, \delta)$$

$$F_k = \begin{cases} 0, & \rho_k \leq l_1 \\ \delta(\rho_k - l_1) \tan \gamma_{new}, & l_1 < \rho_k < l_2 \\ \delta h, & l_2 \leq \rho_k \end{cases}$$

$$\text{or } F_k = \begin{cases} 0, & \rho_k \leq l \\ \delta h, & l \leq \rho_k \end{cases}$$

$$\rho_k = \sqrt{r_{y,k}^2 + r_{z,k}^2}, \quad \tan \gamma_{new} = \frac{h}{l_2 - l_1}$$

[Updated Thrust & Obstacle Constraints]

Algorithm 2 Elevation-angle-based convex optimization algorithm with enhanced obstacle avoidance for powered descent guidance.

```

1: Define tolerance  $\varepsilon_\zeta > 0$ . Define homotopy iteration count  $H \in \mathbb{Z}^+$ .
2: Define iteration count  $j = 0$ . Define Endflag=false.
3: Using Algorithm 1 to calculate the feasible area as shown in Fig. 4.
4: Chose the fuel mass  $m_{fuel}$  and terminal time  $t_f$  in the feasible area.
5: Initial trajectory: solve Problem 6 with selected  $m_{fuel}, t_f$  and  $\gamma_{gs} = 0$ 
   to obtain  $\tilde{u}(\cdot), \tilde{\sigma}(\cdot), \tilde{X}(\cdot)$ .
6: while Endflag=false do
7:    $j = j + 1$ . Assign  $\delta = \min(j, H) / H$ .
8:   Calculate  $F(\tilde{X}(t_k), \delta)$  according to Eq. (32) or (33).
9:   Assign  $z_l(t_k) = \tilde{z}(t_k), z_u(t_k) = \tilde{z}(t_k)$  according to Eq. (30).
10:  Solve Problem 7. Denote the solution as  $\hat{u}(\cdot), \hat{\sigma}(\cdot), \hat{X}(\cdot)$ .
11:  Compute the norm of the slack variable  $\|\zeta(\cdot)\|_2$ .
12:  Check whether the obtained solution satisfies all constraints.
13:  if All constraints satisfied and  $\|\zeta(\cdot)\|_2 \leq \varepsilon_\zeta$  and  $j \geq (H + 1)$  then
14:    Endflag=true.
15:  else
16:    Set the solution  $\hat{u}(\cdot), \hat{\sigma}(\cdot), \hat{X}(\cdot)$  as the initial trajectory. Assign
     $\tilde{u}(\cdot) = \hat{u}(\cdot), \tilde{\sigma}(\cdot) = \hat{\sigma}(\cdot), \tilde{X}(\cdot) = \hat{X}(\cdot)$ .
17:    Endflag=false.
18:  end if
19: end while
20: Assign  $u^*(\cdot) = \hat{u}(\cdot), \sigma^*(\cdot) = \hat{\sigma}(\cdot), X^*(\cdot) = \hat{X}(\cdot)$ .
21: return  $u^*(\cdot), \sigma^*(\cdot), X^*(\cdot)$  as the fuel optimal solution with en-
    hanced obstacle avoidance.
    
```

0) Algorithm 1로 feasible fuel-time area 계산
 m_{fuel}, t_f 선택

1) 선택한 m_{fuel}, t_f 에서 초기 궤적 생성
: Problem 6

2) Taylor / Homotopy update / Slack relaxation
: Problem 7

3) 수렴 판정 및 반복 갱신
: Algorithm 1과 동일 / 선택한 m_{fuel}, t_f 를 유지한 채
Problem 7을 반복 해결

4) 선택한 fuel - time에 대한 수렴 궤적 반환

■ Elevation-angle integral 기반 terminal-time selection

- Algorithm 2는 선택한 m_{fuel}, t_f 에 대해 $u^*(\cdot), \sigma^*(\cdot), X^*(\cdot)$ 를 반환 $\rightarrow$ trajectory $X^*(\cdot)$ 를 생성
- Elevation angle은 시간이 아니라 하강한 고도에 대해 누적 ($d(-r_x) = (-v_x)dt$, 고도 증가 방향)
- 각 Ψ 를 비교하여, 주어진 fuel budget에서 Ψ 가 최대인 t_f 를 최종 선택

$$(m_{\text{fuel}}, t_f) \xrightarrow{\text{Algorithm 2}} X^*(\cdot) \rightarrow r^*(t_k) \rightarrow \theta_k \rightarrow \Psi$$

$$\theta_k = \arcsin\left(\frac{r_x(t_k)}{\sqrt{r_x^2(t_k) + r_y^2(t_k) + r_z^2(t_k)}}\right)$$

$$\Psi = \sum_{k=1}^N \theta_k (-v_x(t_k)) \Delta t = \int_{t_0}^{t_f} \theta(t) (-v_x(t)) dt$$

* θ_k : 6쪽에서 정의한 elevation angle

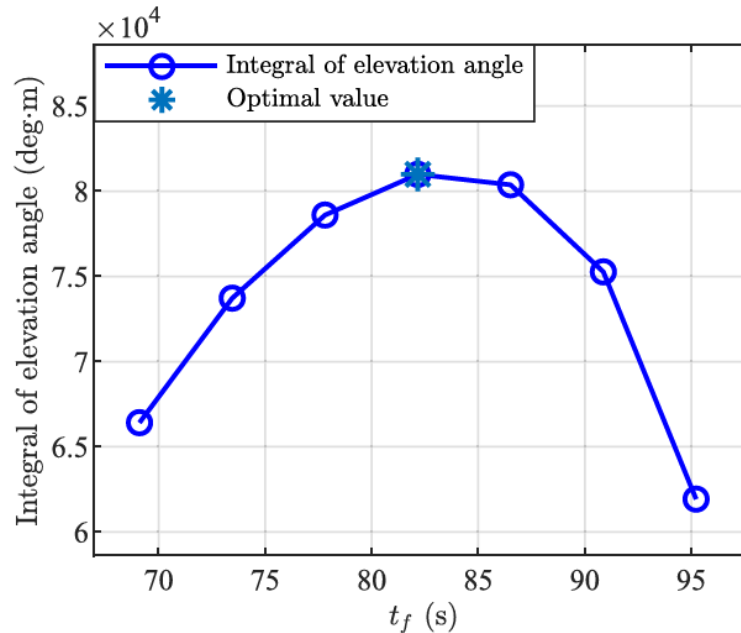


Fig. 15 : 도시된 후보 중 Ψ 가 최대
: $t_f \approx 82.16$ s

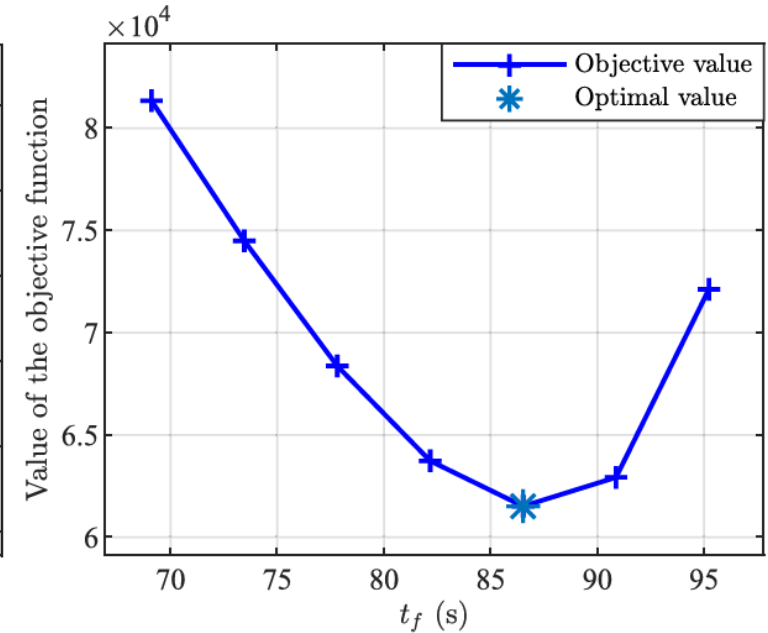


Fig. 16 : 도시된 후보 중 objective가 최소
: $t_f \approx 86.52$ s

Algorithm 1 : Taylor 갱신의 효과

- 고정 Taylor 전개점에서는 terminal time이 길어질수록 말기 추력 profile의 근사 차이가 커짐
- 이전 log-mass trajectory로 전개점을 갱신하면 max - min thrust profile에 더 가까운 해를 얻음
- 비교 조건: conventional glide-slope, $\gamma_{gs} = 10^\circ$, $t_f = 78, 98, 118, 138$ s

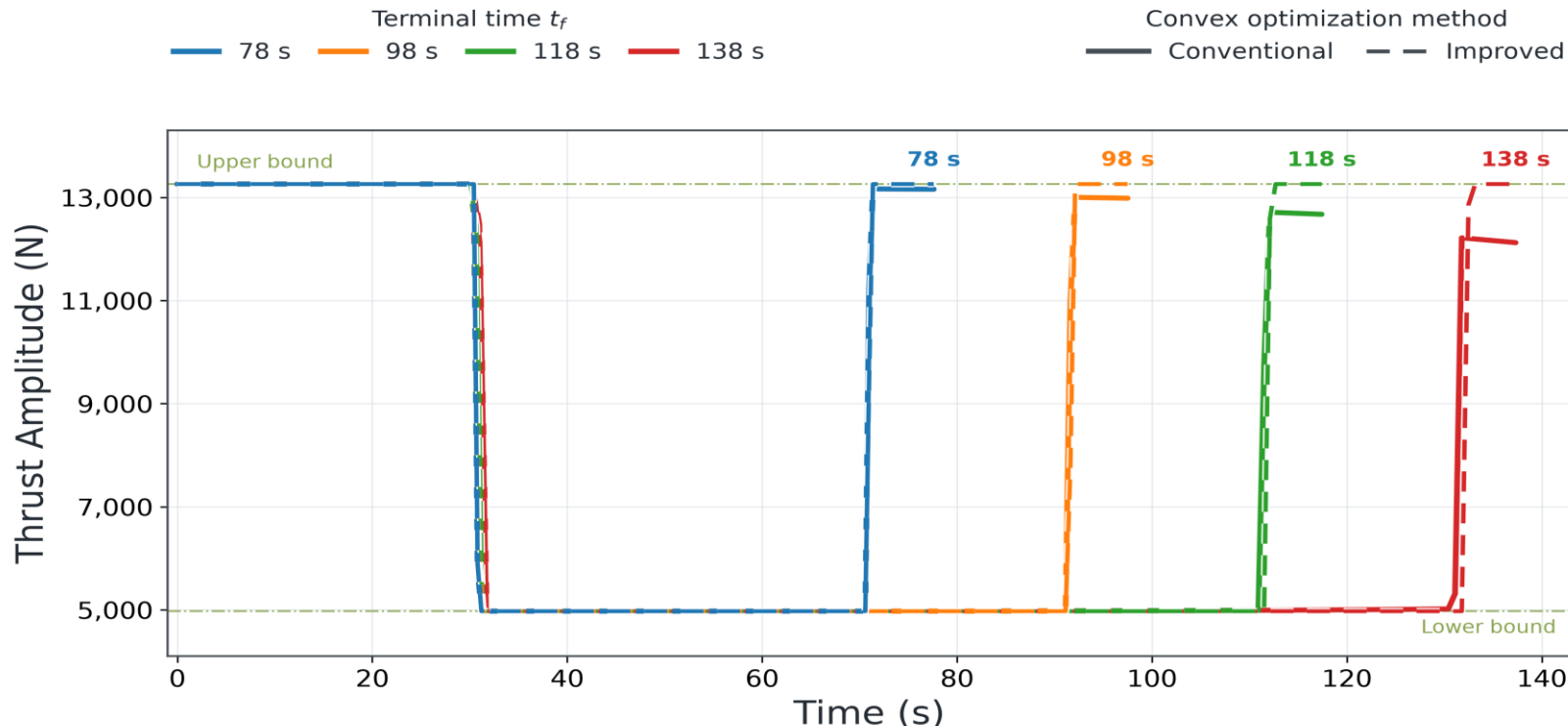


Fig. 18(a) (b) (c) (d) : Thrust comparison under varying terminal times for conventional and improved convex optimization methods.

Algorithm 1 : Computational performance

- Fig. 18(d) 비교에서 Algorithm 1의 평균 CPU time은 0.06 s, GPOPS는 1.12 s
- 비슷한 연산 시간 대비 Algorithm 1뿐만 아닌 FOV 확보를 위한 Algorithm 2도 활용 가능

Table 3

Performance metric of different methods.

Performance metric	Proposed method (Alg. 1)	Proposed method (Alg. 2)	Conventional convex method [20,24]	GPOPS software [47]
Average CPU time (s) (Scenario:Fig. 18 d)	0.06	0.06	0.03	1.12
Fuel consumption (kg) (Scenario:Fig. 18 d)	505.49	505.49	505.51	505.484
Step-shaped obstacle avoidance	Yes	Yes	No	Yes
Improve field of view	No	Yes	No	Yes

Table 3 : Fig. 18(d)의 계산 성능 비교
Conventional glide-slope, $t_f = 138s$

Algorithm 2 : 동일 연료에서 obstacle clearance 개선

- Algorithm 1의 fuel-optimal baseline은 obstacle boundary에 근접할 수 있음
- 동일 연료 409.12 kg에서 terminal time을 변경하면, 회피 여유와 말기 수직 접근 형상이 개선된 후보 궤적을 얻음

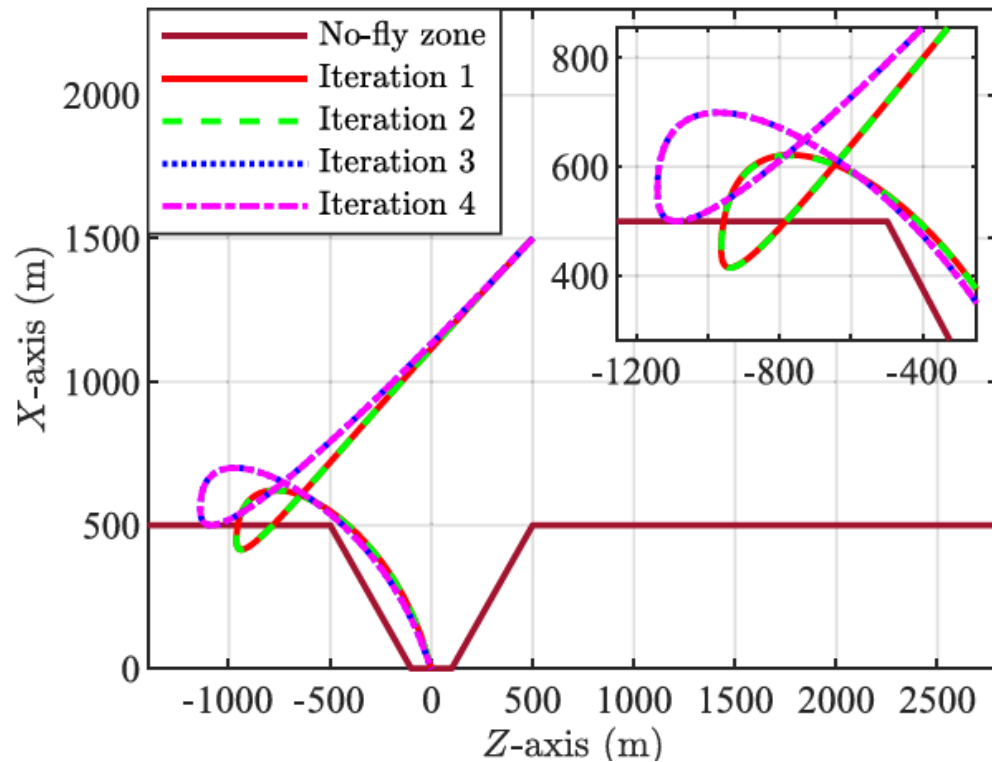


Fig. 21 : Algorithm 1 - 최종 해: Iteration 4
 $t_f = 100s, Fuel = 409.12kg$

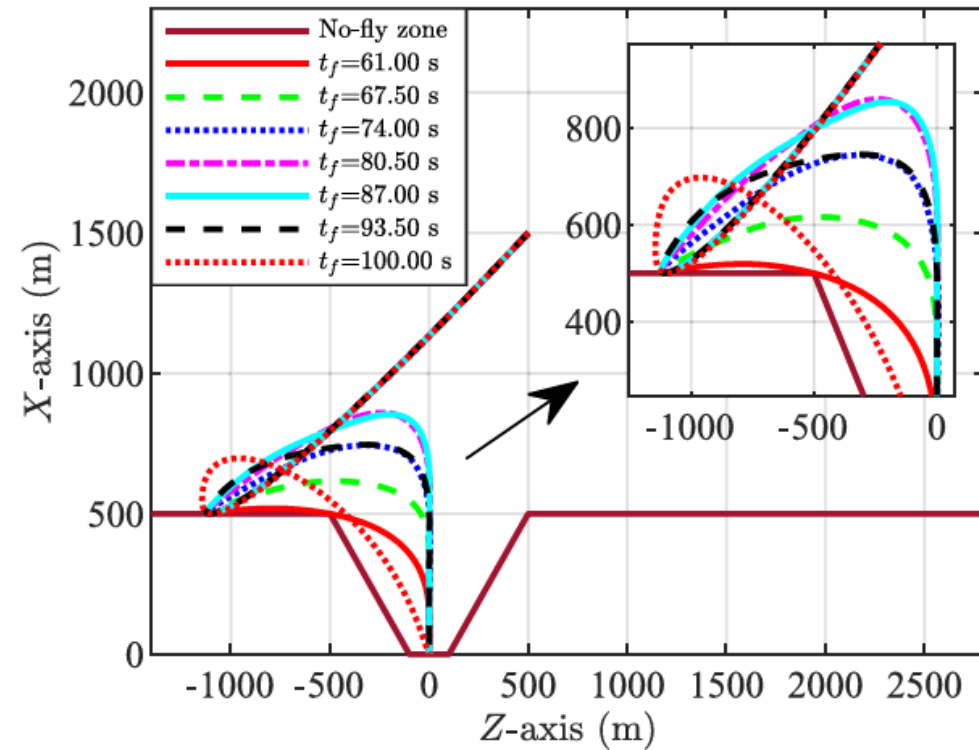
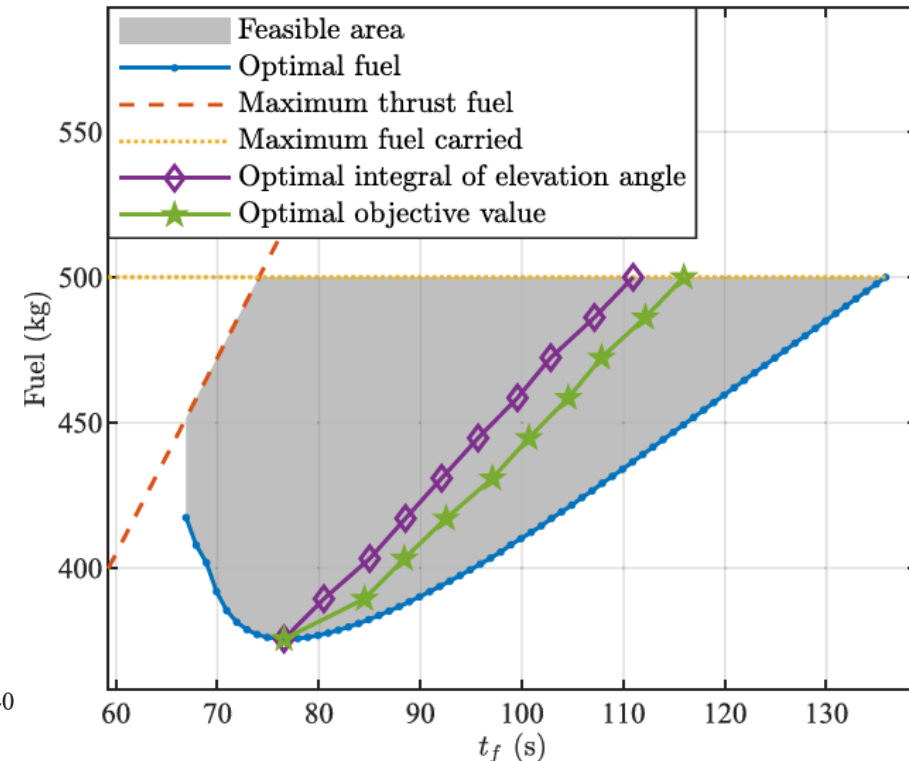
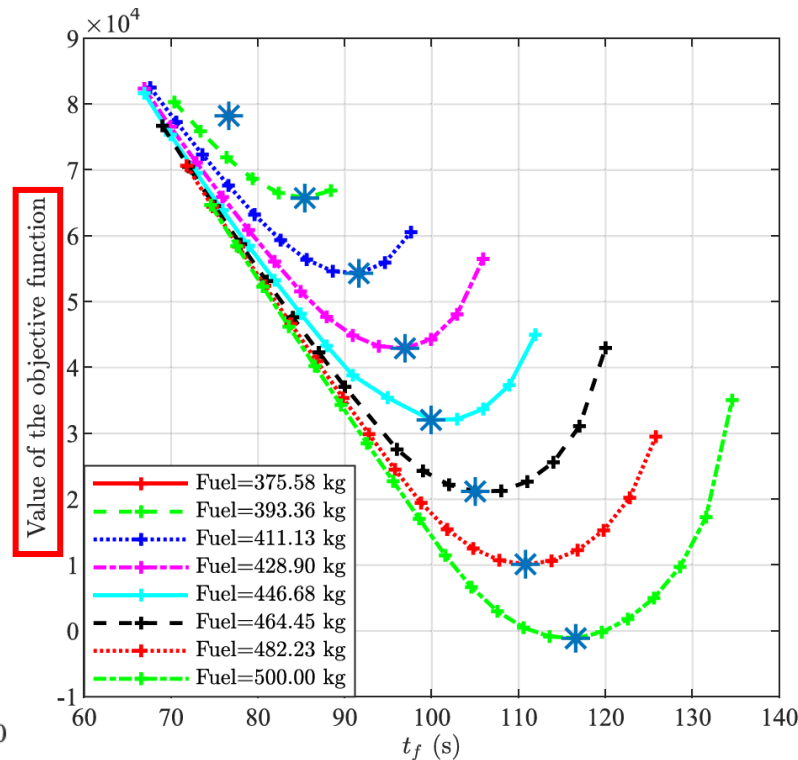
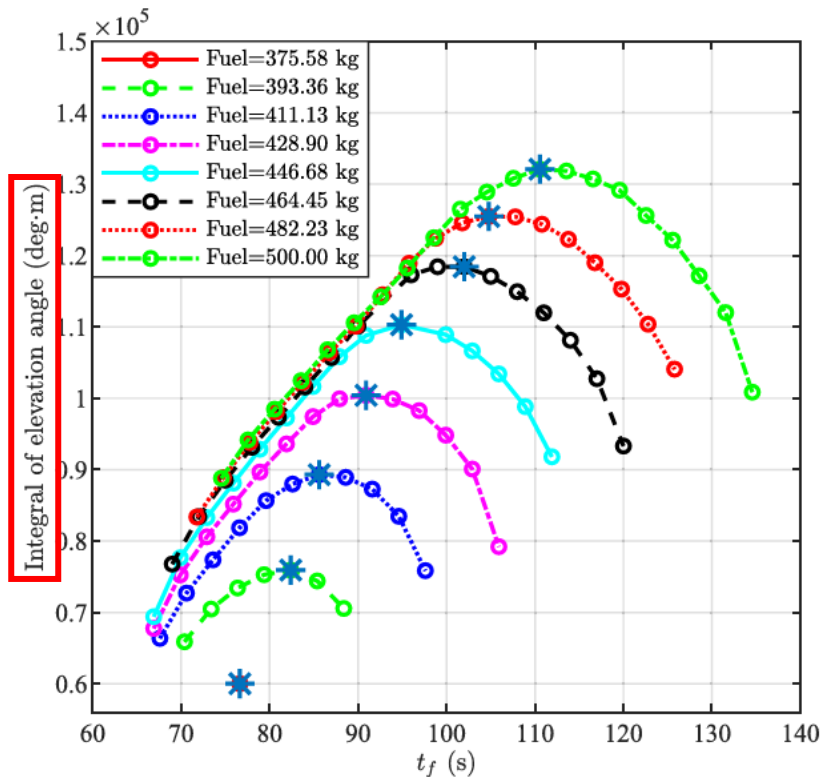


Fig. 27 : Algorithm 2 - terminal 후보 궤적
 $t_f = 61-100s, Fuel = 409.12kg$

Optimal guidance : fuel budget별 terminal-time 선택

- 제시된 연료 수준별 결과에서, Ψ 가 최대인 terminal-time 후보를 선택
- 선택된 (t_f^*, m_{fuel}) 를 연결하여 feasible area의 Ψ -optimal curve를 구성
- 보라색은 Ψ 기준 선택, 초록색은 position objective 기준 선택, 파란색은 minimum-fuel boundary



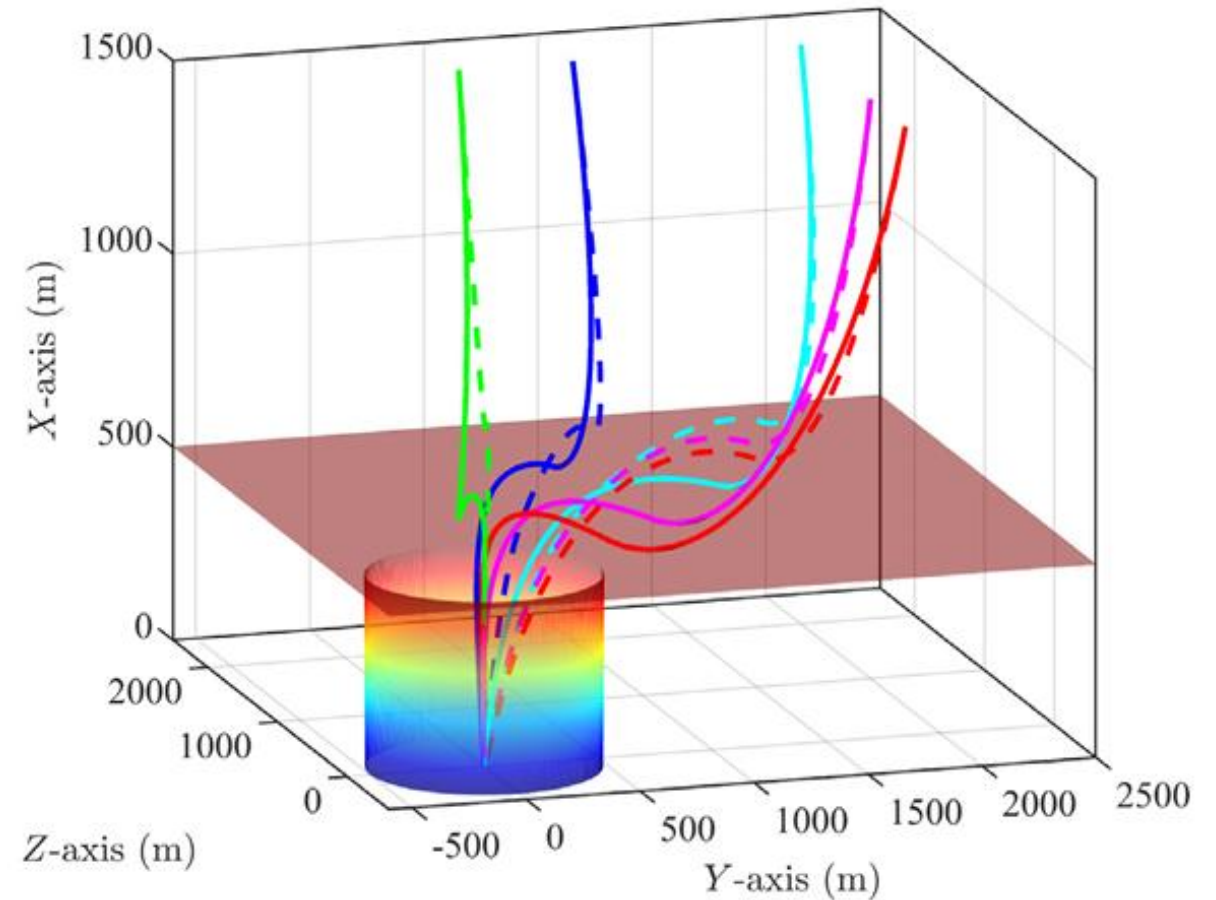
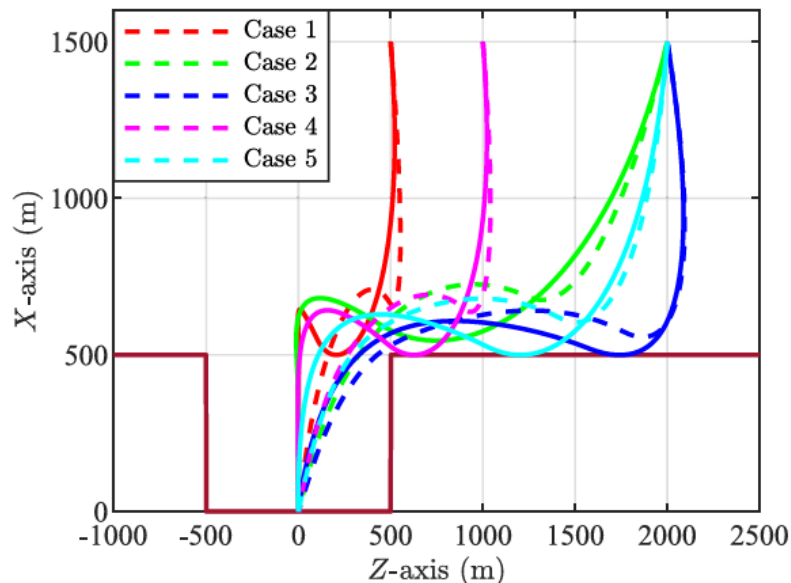
■ 3-DoF 적용 : 초기조건 변화에 따른 결과

- 점선: Algorithm 1, 실선: elevation-angle-based optimal guidance
- 동일 연료에서 obstacle avoidance를 유지하면서, 착륙 직전의 수평 이동을 줄여 near-vertical approach를 형성

Table 5

Initial conditions for 3-DoF cases.

Cases	Initial position $\mathbf{r}_0$ (m)	Initial velocity $\mathbf{v}_0$ (m/s)	Terminal time t_f (s)
Case 1	$[1500, 2000, 500]^T$	$[-75, -20, 10]^T$	80
Case 2	$[1500, 500, 2000]^T$	$[-75, 0, -30]^T$	80
Case 3	$[1500, 1000, 2000]^T$	$[-75, 20, 20]^T$	80
Case 4	$[1500, 2000, 1000]^T$	$[-75, -10, 10]^T$	80
Case 5	$[1500, 2000, 2000]^T$	$[-75, 10, -10]^T$	80



■ Conclusion

- Algorithm 1: homotopy obstacle handling과 Taylor 갱신으로 **fuel-optimal baseline** 구성
- Algorithm 2: selected fuel budget 안에서 **vertical-biased trajectory** 생성
- 전체 framework: elevation-angle integral Ψ 로 terminal-time을 선택
- 제시된 실험에서는 **동일 연료로 obstacle clearance와 terminal viewing geometry를 개선**

■ Future work

- 3-DoF 수치 사례의 결과이며, 실제 camera / touchdown dynamics 검증과는 구분
- near-vertical landing property를 유지하는 6-DoF powered descent로 확장

THANK YOU

MAIL lewis430@inha.edu

TEL 000-000-0000

CELL +82 10-6590-0853



인하대학교
INHA UNIVERSITY



항공우주제어시스템연구실
Aerospace Control & Systems Laboratory